Note

- Use presentation format provided by RPS and save it in your local machine and also upload it in the GitLab as PDF file.
- Rename all the files with the naming convention as “**Capstone-25SUB5011-<5-digit-ID>”** as PDF document and upload it on GitLab.
- Upload only PDF documents with the specific name as mentioned above.
- This document contains **8 projects** in it, each in a different page.
- Its compulsory to attempt 4 projects.
- Last project (Project 8) is compulsory for all the participants.
- Use existing VM ONLY if its asked in the question, else VM must remain powered-off.
- Project will NOT be considered, if current date and time within each VM is not visible at the right-bottom corner of the screenshot.

Project Task 1: - "Establishing domain environment."

Objective: Create a domain controller using domain name as "capstone.org" and domain join Member01 after installation. Add both VMs under LAN Segment (named **capstone-lab**) and provide the IP address under "Class C" range and take the snapshot of the VMs after installation and verify.

Setup:

- ✓ Use VMWare workstation 17 or above to create a windows server 2022 as domain controller with a custom LAN Segment names as "**capstone-lab**".
- ✓ Provide static IP as 192.168.10.10/24 with loopback DNS address & perform post-installation configuration.
- ✓ Install the required role and promote the server to domain controller.
- ✓ On DC, create a user named "dcadmin" with the 'Administrator-level' permissions.
- ✓ Install another windows server 2022 with the name "Member01" domain join to domain controller & verify.
- ✓ Add additional 25GB HDD to the domain controller machine and mount it as NTFS file system.
- ✓ Create a powershell script that monitors the following application and generated alert/logs when these applications are opened.
    - Powershell.exe
    - Cmd.exe
    - Notepad.exe
    - Mstsc
    - Wf.msc
    - Firewall.cpl
    - Compmgmt.msc
- ✓ Write PowerShell cmdlet to list latest 5 event logs which displays full/detailed logs (without truncating the output).

Project deliverables:

- ✓ Document (take screenshot) installation of Domain Controller and Member01 machine.
- ✓ Document (take screenshot) the post-installation setup & Domain Controller promotion.
- ✓ Document (take screenshot) for creating user equivalent to Domain Administrator.
- ✓ Document (take screenshot) addition of extra 25GB HDD after installation and promotion on DC.
- ✓ Document (take screenshot) powershell script and its output that shows alerts were captured in a notepad file.
- ✓ Document (take screenshot) powershell cmdlet to list event logs.
- ✓ Document (take screenshot) for pricing calculator and calculate the cost of all the VMs/Storage and other resources present in this question and display the amount in INR only.

Project Task 2: - "Docker with CentOS"

Objective: Install CentOS Stream 9 operating system using standard pre-requisites and name it "Cent9server" and take the snapshot of the VM after installation.

Setup:

- ✓ Use VMWare workstation 17 or above to create a CentOS 9 OS and with one network adapter set to a custom LAN Segment names as "**capstone-lab**" and another network adapter set to NAT.
- ✓ Provide static IP as 192.168.10.11/24 to the network adapter in LAN Segment (capstone-lab) and ensure both network adapters are active and have IP addresses.
- ✓ Ping to google.com from the terminal.
- ✓ Then install docker from scratch and pull custom image "***<u>tomarj44/c7custom:v3</u>***" from DockerHub using appropriate command and verify of the image is present in the system.
- ✓ Add the following ports and services in the firewall and verify:
  - o Add ports: 80, 22, 53, 3128
  - o Add services: NTP, Samba
  - o Create a rule to block ICMP packets and verify.
  - o Create a rule to block FTP service and verify.
- ✓ Gather the recent logs for all the steps executed till now.
- ✓ Use existing windows machine and connect to this Linux machine (Windows → Linux) via SSH on powershell terminal.

Project deliverables:

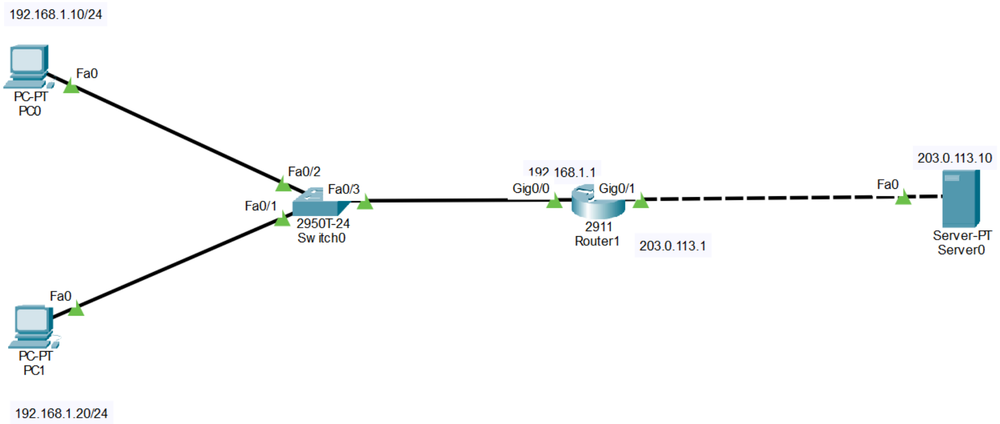
- ✓ Document (take screenshot) of CentOS installation and adding static IP address.
- ✓ Document (take screenshot) of adding extra NIC card under NAT.
- ✓ Document (take screenshot) of docker commands and docker image(s) present within linux.
- ✓ Document (take screenshot) linux command to list logs on terminal.
- ✓ Document (take screenshot) of the firewall commands for adding, removing and displaying docker images.
- ✓ Document (take screenshot) of connecting windows to linux using powershell.
- ✓ Document (take screenshot) for pricing calculator and calculate the cost of all the VMs/Storage and other resources present in this question and display the amount in INR only.

Project Task 3: - “Building network with Cisco Packet Tracer”

Objective: Use existing Cisco packet tracer (CPT) setup either on your base laptop or RPS lab.

Setup:

- ✓ **CPT-File 1**: Add 2 Laptops and a switch (of your choice) in CPTs workspace and connect them with the appropriate cable. Provide then static IP address and both Laptops using built-in command prompt and verify if its pinging and close this workspace.
- ✓ **CPT-File 2**: Now, create another workspace and add 2PCs, 1 switch and 1 server and provide IP address to this setup using DHCP and test if they are pinging each other.
- ✓ **CPT-File 3**: add another file and perform Network Address Translation (NAT) for the following setup.



Project deliverables:

- ✓ Document (take screenshot) of CPT-File-1 with IP address and other required components.
- ✓ Document (take screenshot) of CPT-File-2 with IP address and other required components.
- ✓ Document (take screenshot) of CPT-File-3 with IP address and other required components.

Project Task 4: - “Blocking websites using Palo-Alto Firewall”

Objective: Import a fresh Palo-Alto (PV) firewall and login to it after resetting its old password.

Setup:

- ✓ Import Palo-Alto Firewall and connect to it using CLI and change its password and access its management interface IP address using appropriate command.
- ✓ Login to Palo-Alto using management interface IP address from the web browser.
- ✓ Create rules for block the following websites using AppID:
    - o Facebook
    - o YouTube
- ✓ Create another rule to perform URL Filtering based on the category:
    - o Hacking → Block
    - o Gambling → Block
    - o Adult → Block
    - o Malware → Block
    - o Phishing → Block

Project deliverables:

- ✓ Document (take screenshot) import the PA firewall.
- ✓ Document (take screenshot) resetting the password.
- ✓ Document (take screenshot) accessing the PA using web browser.
- ✓ Document (take screenshot) of blocking websites using AppID.
- ✓ Document (take screenshot) of performing URL filtering.

Project Task 5: - “Kali linux setup”

Objective: Import a fresh kali linux machine use network adapter as NAT and ping to google.com and verify.

Setup:

- ✓ Import a new kali linux and test if internet is reachable or not.
- ✓ Perform a brute force attack for the user named “captoneadmin” on the DC machine. You can use existing DC for this task. Use password file to perform attack.
- ✓ Create a malware test file, copy this file to windows machine and scan it using the defender anti-virus and see if it’s getting notified by the AV.

Project deliverables:

- ✓ Document (take screenshot) importing kali linux.
- ✓ Document (take screenshot) creating a user on DC.
- ✓ Document (take screenshot) for performing brute force attack.
- ✓ Document (take screenshot) to test the malware on the DC machine with the anti-virus.

Project Task 6: - "CIS Benchmark"

Objective: Install a Windows 10 Pro edition and configure CIS benchmark standards.

Setup:

- ✓ Install Windows 10 Pro and configure IP address. Configure this Windows 10 machine as per standards of CIS as given below.
- ✓ Enable "Administrator" on Windows 10.
- ✓ Configure password policy:
    - Enforce password history: 24
    - Max. password age: 60 days
    - Min. password age: 1 days
    - Min. password length: 12
    - Complexity requirement: Enabled
- ✓ Configure lock out policy:
    - Account lockout threshold: 5 attempts
    - Lock duration: 15min
    - Reset counter after: 15min
- ✓ Configure security policy:
    - Set interactive logon message title: "Welcome to Capstone Project"
    - Set interactive logon message text: "This is Capstone project text title by <Your Name>"
- ✓ And verify all the policies.

Project deliverables:

- ✓ Document (take screenshot) installing Windows 10
- ✓ Document (take screenshot) details of password policy.
- ✓ Document (take screenshot) details of account lockout policy.
- ✓ Document (take screenshot) details of security policy.

Project Task 7: - "Infection Monkey"

Objective: Gather security logs using Infection Monkey.

Setup:

- ✓ Download, install and run Infection Monkey on a CentOS 9.
- ✓ Configure Infection Monkey and run it for the security report.
- ✓ Scan the complete network and wait until the scan is complete and view the report.
- ✓ View the map of connected devices.

Project deliverables:

- ✓ Document (take screenshot) downloading and installing Infection Monkey.
- ✓ Document (take screenshot) scanning the network.
- ✓ Document (take screenshot) report analysis of the scanned devices within the network.

Project Task 8: - “Mandatory Project”

Objective: Use existing Domain controller with linux server and kali linux and perform DNS Spoofing and redirect the user to fake (phished) google web page using credential harvesting technique. Use Linux machine to configure Squid proxy and monitor the whole network (you can use NAT network or LAN segment, according to your need). And block facebook.com, youtube.com, Instagram.com and verify it on the DC machine. Take this monitoring output of squid proxy server from terminal and save it in a file and view this on linux machine.

Use CyberChef and create Base64Encoded value for the cmdlet ‘**write-host “Welcome to Capstone Project”**’ and execute it on DC machine and verify. Also check for the relevant log for the encoded command using powershell script.

Setup:

- ✓ Use Centos 9, DC, Kali linux and CyberChef for this task.
- ✓ Perform ARP poisoning attack on DC and Kali linux and fetch the credentials and display it on Kali linux terminal.
- ✓ Use Squid proxy server to monitor and block the given websites & verify it in DC.
    - o Facebook.com
    - o YouTube.com
    - o Instagram.com
- ✓ Access the squid proxy logs and save them in another file on root’s home directory and verify.
- ✓ Use DC to access CyberChef and create an encoded value for the given string and execute on DC and verify the output and display the log using powershell script.

Project deliverables:

- ✓ Document (take screenshot) performing ARP poisoning attack and fetching credentials.
- ✓ Document (take screenshot) if installing Squid proxy and blocking the gives sites and capturing/copying logs in another file.
- ✓ Document (take screenshot) converting the string in encoded format and then executing it using required “powershell -enc” command.